AKSHAY R

Full-Stack Developer — Python, Django, React.js, MySQL — AI Application Engineering

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SUMMARY

Focused on backend-based software engineering using Python, Django, and Django REST Framework, along with AI-powered applications. I have hands-on experience building RAG applications, OCR pipelines, NLP solutions, local LLM integrations using Ollama, and full-stack platforms. I also have experience working on freelance projects based on client requirements. Strong in authentication, role-based workflows, semantic retrieval, data validation, PostgreSQL, Docker, and service-layer architecture.

SKILLS

Languages: Python, JavaScript, SQL, C **Backend:** Django, DRF, FastAPI, REST APIs, JWT, Role Based Access Control
Frontend: React.js, HTML5, CSS3 ,Tailwind css **Databases:** PostgreSQL, MySQL, MongoDB, Firebase,Supabase, Firestore, ChromaDB
AI/ML: Machine Learning,Fine Tuning, NLP, RAG, Semantic Search, TF-IDF, Cosine Similarity, LLM Integration
AI/Document: Ollama, Gemma 3, Llama 3, PyTorch, Transformers, Tesseract OCR, PyTesseract, OpenCV, PyPDF
Tools: Docker,Git, GitHub, Vercel, REST API Testing(Postman)

PROJECTS

Janseva AI – AI-Powered Citizen Complaint Intelligence Platform GitHub

Python, Django, DRF, React.js, Tailwind CSS, Ollama, Tesseract OCR, PyTorch, Transformers, OpenCV, PyPDF

- Built an **AI-powered citizen complaint platform** that converts unstructured letters, PDFs, and images into structured complaint records.
- Developed an end-to-end pipeline: **document extraction → OCR → text analysis → LLM-based extraction → validation → complaint processing**.
- Integrated **Ollama** to extract complaint details and identify missing information from OCR-processed text.
- Implemented **validation, controlled output, and normalization** for complaint categories, priorities, and other structured fields, along with duplicate complaint detection.
- Implemented a **Telegram bot** for complaint submission and status tracking using complaint ID and phone number.
- Added support for **English and Malayalam complaint processing** using Tesseract OCR and developed the complete backend with **Python/Django** and frontend with **React.js/Tailwind CSS**.

Doctor AI – Retrieval-Augmented Document Intelligence Platform GitHub

Django, DRF, React.js, Axios, ChromaDB, Sentence Transformers, Ollama, Llama 3, Docker

- Built a platform where authenticated users can upload documents and ask questions based on the uploaded content.
- Implemented the complete RAG pipeline: **document upload → text extraction → chunking → embeddings → vector storage (ChromaDB) → semantic retrieval → LLM generation**.
- Used **Sentence Transformers and ChromaDB** for generating semantic embeddings and storing vectors for efficient retrieval of relevant document context.
- Integrated the **Llama 3 model via Ollama** to generate context-aware responses based on the uploaded documents.
- Implemented **JWT authentication and user-level document isolation** using Django REST Framework.
- Developed the frontend using **React.js and Axios** for API communication and document interaction.
- Containerized both the **frontend and backend using Docker** for consistent development and deployment.

AI-Powered Smart Interview Platform GitHub

React, Node.js, Express.js, Python, NLP, OpenAI API

- Developed an AI-assisted interview platform combining **full-stack architecture, NLP-based answer evaluation, and LLM-powered question generation**.
- Implemented automated evaluation using **TF-IDF, cosine similarity, keyword extraction, and semantic matching** to compare candidate responses with expected concepts.
- Integrated an **LLM API for dynamic question generation and contextual feedback**, supporting adaptive interview workflows.
- Designed REST APIs for **interview sessions, questions, candidate responses, evaluation workflows, and performance analysis**.

Emergency Service – Role-Based Service Request Platform GitHub

Django, DRF, React, PostgreSQL, Docker, JWT

- Developed and deployed a **role-based service platform** connecting customers with providers across multiple emergency-service categories.
- Designed REST APIs for **registration, service requests, category filtering, provider assignment, status transitions, completion tracking, and reviews**.
- Implemented **JWT authentication, role-based workflows, provider job tracking, rating aggregation, and review validation** using DRF and PostgreSQL.
- Containerized the backend and deployed the application using a separated **frontend/backend architecture**.

EDUCATION

B.Tech in Computer Science Engineering – College of Engineering Kidangoor 2022–2026
Class X: 91 — **Class XII:** 96

CERTIFICATIONS

Artificial Intelligence Internship – Qnayds LLP — **Python Django with Gen AI – HACA** Jun–Dec 2026